LIS 467 Web Development & Information Architecture

Assignment 1: Project Plan & Needs Assessment

1. Company Name: Agentic Curriculum Designers, Inc.

Logo:



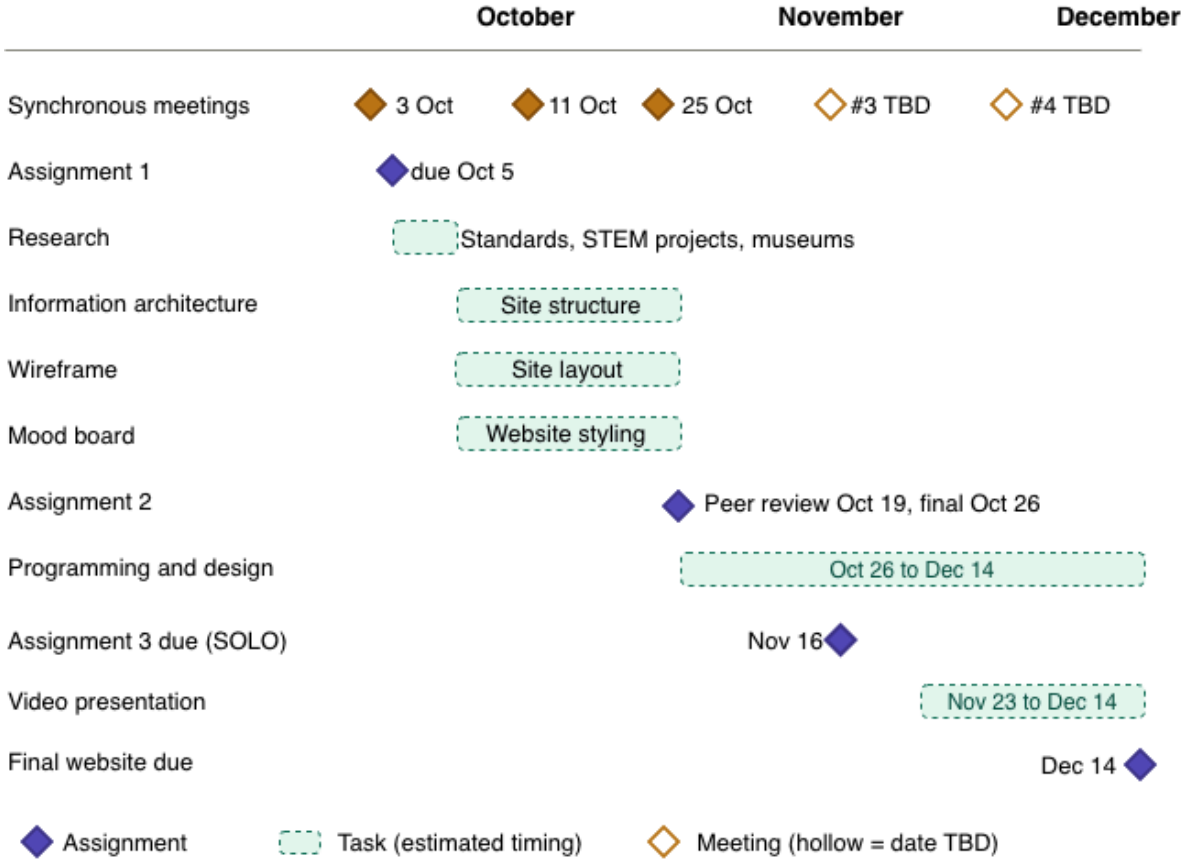
2. Team members and roles (when working alone, imagine you're a web design company of 3-5 people; you can use fictitious names for other team members):

<Ambrose Choyce> - Designer
<Kameron Cole> - Programmer
<Jayne Sobczyk-Moore> - Designer

Reasons for choosing the number of members and assigning the roles this way:

We felt that three members was the perfect amount of people needed for a project of this scale and type. For our roles, we had discussed when initially signing up what roles each of us felt would fit us best based on our individual skill sets, and these reflect our own choices. Kameron has had a lot of experience with programming projects in the past, and both Ambrose and Jayne are more versed in the visual aspect of web design.

3. Project schedule / timeline



4. Skills needed – and plans to acquire them

1. AI integration into interactive teacher/student webpage
    a. **Acquisition:** Research this topic through articles online and reviewing sample AIs similar to the one we plan to create.
2. Similarity/semantic search for webpages, with/without a database
    a. **Acquisition:** Research this topic through articles online and reviewing sample semantic searches similar to the one we plan to implement.
3. HTML5
    a. **Acquisition:** Research this topic through articles and other teaching resources available online.
4. Node.js
    a. **Acquisition:** Research this topic through articles and other teaching resources available online.
5. Javascript
    a. **Acquisition:** Research this topic through articles and other teaching resources available online.

5. Site chosen – reasons for choice, and if you'll get in touch

The site we chose for this assignment is the Museum of Science (MOS) in Boston (https://www.mos.org/). The current MOS website lacks important background information on how their exhibits relate to specific STEM topics, nor how they facilitate learning connections for teachers, parents, or students. In order to create a unified picture of how a school assignment might incorporate an exhibit's teachings, we plan to connect teaching standards to exhibit information and external resources for educators. For students, exhibit information will be connected to external resources, while additional STEM activities will be connected to exhibit information for parents and other visitors. Additionally, we hope to incorporate a customized AI chatbot based on the user's identity that will help them navigate and search the site easier. Once the project is completed, we hope to share it with the museum.

6. Audience / stakeholders

The MOS needs to attract interest from local families, and educators interested in the teaching materials they sell online as well as potential field trips. Educators need easy ways to compare lesson plans and activities with state mandated learning objectives. Fielding complicated questions from educators is time consuming and making the website more user friendly will help staff as well.

As such, the audience for this website will be teachers and parents who hope to teach their students or children more about STEM topics and wish to use the MOS for this purpose. An additional audience will be students doing research on STEM topics, either out of their own desire to learn or based on assignments given to them by their teachers/parents. The website will link additional resources about the exhibits and assist users in topic exploration.

7. Plan to gather requirements

1) Need to collect and download the https://www.doe.mass.edu/stem/ste/standards.html.
2) Investigate the interactive DESE Standards Navigator Tool.
3) May need to interview public school STEM teachers for design requirements.
4) Curate document seats geared to the student persona, that would aid in structure research assignments, per activity.

8. Mission/purpose

Mission Statement:

The mission of this website is to facilitate interactive learning and assist both educators and students in making connections with more intuitive searching techniques.

Design Statement:

Provide unity in the design of course curricula and education activities, utilizing the exhibits at the museum, but cross-referencing these with actual educational objectives of the State Massachusetts, particularly in STEM. The idea is: an educator wants to create an assignment that incorporates the museum experience, but aligned with specific STEM Objectives. For this, they would need additional information (the MOS site does not provide much information), a dynamic linking of the exhibit, the Objective, and extra information

resources. Additionally, the creation of a chat bot for students and teachers linked to an AI model for open discussion.

9. Typical users

The typical users for this hypothetical website will be those hoping to learn more about STEM topics, with the different types of users being teachers, parents, and students.

10. Sample persona



Persona: Ms. Frizzle

- Teaches 5th grade science classes at a Boston public elementary school.
- Has been teaching for 15 years, always looking to update and improve her curriculum.
- Her science course has the budget for 2 full-day field trips per year. She must research and present these trips to the school for approval on top of her busy teaching schedule.
    - These field trips must fulfill required learning objectives & have coordinating lesson plans at the time of approval.
    - Field trips must be approved before the beginning of the school year (early September). She needs ample notice of planned exhibits.
- She is always looking for interactive activities for students & new ways to get them interested in learning.

Assignment 2: Information Architecture, Wireframe & Mood Board

1. Site Description

We would like for any visitors to our site to feel that:

a. The whole searching process felt very streamlined and to-the-point.
b. The website was visually appealing and descriptive of current MOS events without being cluttered.
c. The website helped them connect resources with exhibit information easily.

2. Reference Websites

1. Smithsonian Institute Museums: https://www.si.edu/
    a. Has a carousel with upcoming events. We had discussed as a group that, no matter what, a museum needs to have a prominent events carousel.
    b. Has a fully developed educational component, with a STEAM section. It differs from our site such that the Smithsonian has developed its own STEAM curriculum, whereas ours coordinates with Massachusetts public curriculum.
    c. Has a local search interface, meaning it searches within the following subsite: https://ssec.si.edu/. This is almost exactly what we intend for our own MOS site, namely that it will act as a subsite of the main MOS site that is focused on educational resources.
2. The Metropolitan Museum of Art (MET): https://www.metmuseum.org/
    a. No carousel is present, but has a long video as the front matter.
    b. They have an educational component: https://www.metmuseum.org/learn/learning-resources. This is not a subsite per se, but just a subfolder that includes an Educator persona.
    c. Their search features exhibits and newspaper clippings about those exhibits. There is also a site search, as seen in the following JavaScript code:

```javascript
<script type="text/javascript">
var pageConfig = {
searchApp: {
name: 'siteSearch',
id: 'siteSearch'
}
};
```

3. National Gallery of Art: https://www.nga.gov/educational-resources
    a. Instructions are provided for materials and warm-up questions for assignments based on the work for any parents/teachers that wish to continue education on the work. This corresponds with our goal for our website to incorporate Massachusetts public curriculums to enhance teaching opportunities for parents and teachers.
    b. Drop-down menus are provided for more specific searching (i.e. who is searching, what kind of resource is being searched, what is the user's specified grade level). This feature is similar to our goal of creating a chatbot that will help users specify via search terms which items they want to learn about.

3. Use Cases

1. EDUCATOR: An educator needs to fulfill Massachusetts State Standards (MASS STE) using expeditionary learning. They come to the mos.org site, initially just to explore the offerings. Without our site, this "browsing" would be clumsy and frustrating; our

site knows exactly what they want and streamlines the process. The educator simply needs to use our semantic search to state “I am a MA 6th grade science teacher. Which STE standards can I fulfill at MOS?”, and they will be shown the proper exhibits, toolkits, and a mapping of tools to standards. The educator can also navigate our site through custom navigation.

2. STUDENT: A student receives an assignment from their teacher (either a MOS-centric assignment or a general one) that requires them to prepare content, such as an essay. Entering our website, they can start with the semantic search bar, but select the student filter, and be shown exhibits and kits that will let them use museum resources. They will also be shown relevant research documents from the websites’ curated collection. Like educators, they can use the custom navigation as well.
3. PARENT: A parent just wants to buy tickets for a current exhibit. They can use the semantic search bar to find something their kids would like based on their search query (e.g. “My son is into robots and space”). Our site will present the parent with exhibits and ticket options, as well as learning kits to purchase if they want to. Because they chose the parent filter in search, they will not interact with any of the STEs or the museum’s Youth Engineering Solutions.

4. Metadata

1. Museum of Science Boston educator resources
2. Youth Engineering Solutions (YES) curriculum
3. Pre-K to Grade 8 STEM curriculum
4. Massachusetts STE standards aligned lesson plans
5. Engineering design activities for kids
6. Hands-on STEM activities for teachers
7. Museum of Science field trip and exhibit alignment
8. Elementary and middle school engineering units
9. Wee Engineer early childhood STEM
10. Free K–8 STEM lesson plans Massachusetts
    - ★ NOTE: Since our site will have a semantic search with an AI assistant, search terms will not be critical; they can simply enter human sentences which the search will effectively translate.

5. User Scenario

EDUCATOR:

a. Needs to fulfill curriculum requirements either from the State or from their school for STEM.
b. Semantic search, with special filters for educators. The site design will lead them down one of three paths, starting from either 1) STE standards 2) Exhibits 3) Activities; each path contains the other two paths.
c. The goal with semantic search/AI is to remove most friction involving learning, but as with any AI, hallucinations are always a risk.

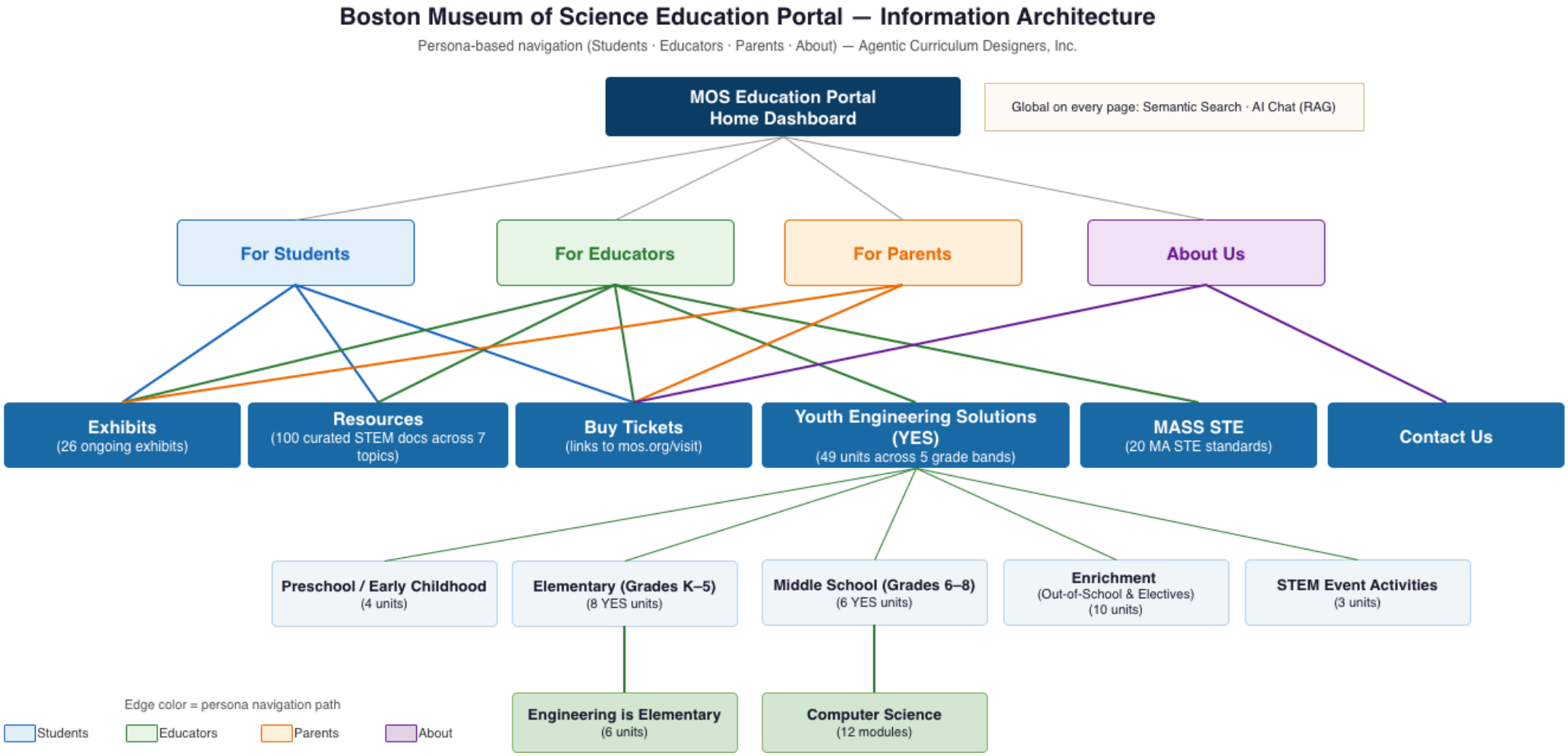
STUDENT:

a. Needs to complete an assigned research essay.
b. Semantic search, with special filters for students. Students will have access to curated research articles, provided by the AI/semantic search, with links to exhibits. They will not be exposed to State standards
c. The goal with semantic search/AI is to remove most friction involving learning, but as with any AI, hallucinations are always a risk. This risk is increased with young children, as we will need to ensure kids are directed to credible and appropriate sources of information.

PARENT:

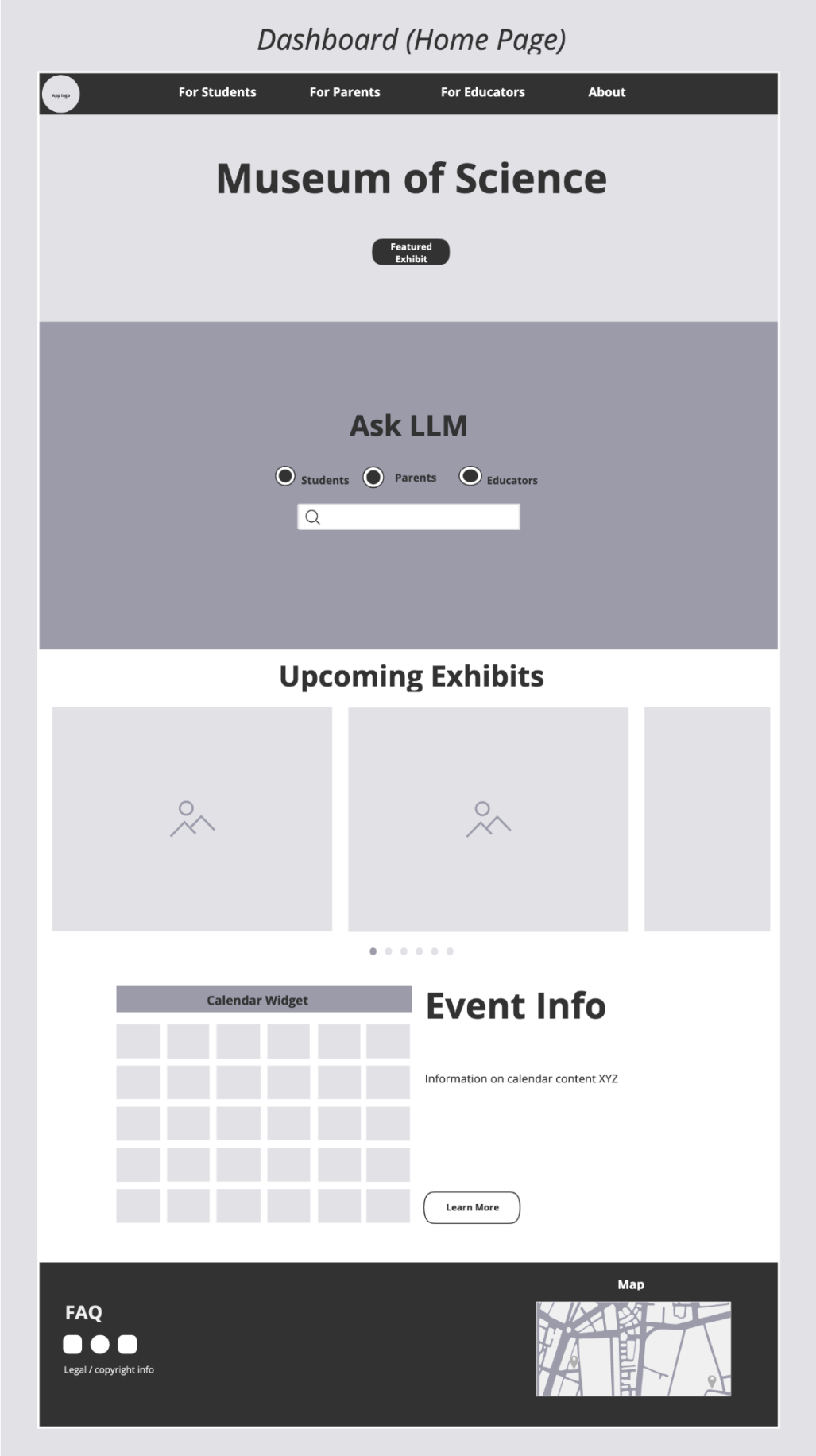
a. Wants to find something educational and fun for their kids
b. Semantic search, with special filters for parents/visitors, such as exhibits. The AI/semantic search will match the exhibits exactly to their child's interests. There will also be ticket options provided. They will not be exposed to State standards
c. The goal with semantic search/AI is to remove most friction involving learning, but as with any AI, hallucinations are always a risk. Navigation controls tailor their information seeking path.

6. Information Architecture



***Additionally, please see the submitted excel sheet.**

7. Wireframe (image and/or link)



Link to Dashboard

Educator Search Result Page

App logo | For Students | For Parents | For Educators | About

Search Results

Query Restated

LLM output: lorem ipsum dolor sit amet consectetur adipiscing elit quis laborum cum ipsum dolore excepturi eum vero quo

Relevent Exhibit

lorem ipsum dolor sit amet consectetur adipiscing elit quis laborum cum ipsum dolore excepturi eum vero quo

MASS STE
MASS STE

More



Relevent Exhibit

lorem ipsum dolor sit amet consectetur adipiscing elit quis laborum cum ipsum dolore excepturi eum vero quo

MASS STE
MASS STE

More



Relevent Exhibit

lorem ipsum dolor sit amet consectetur adipiscing elit quis laborum cum ipsum dolore excepturi eum vero quo

MASS STE
MASS STE

More



Ask LLM

Students | Educators | Caretakers

Resource Results

PDF **Document Title**
LLM output: lorem ipsum dolor sit amet consectetur adipiscing elit quis laborum

PDF **Document Title**
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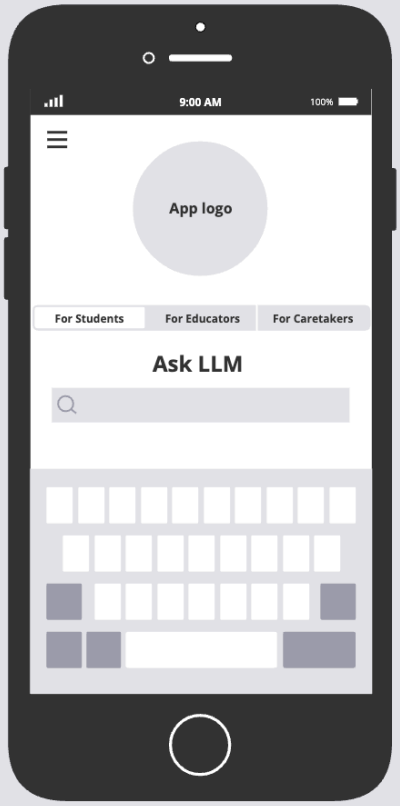
FAQ

Legal / copyright info

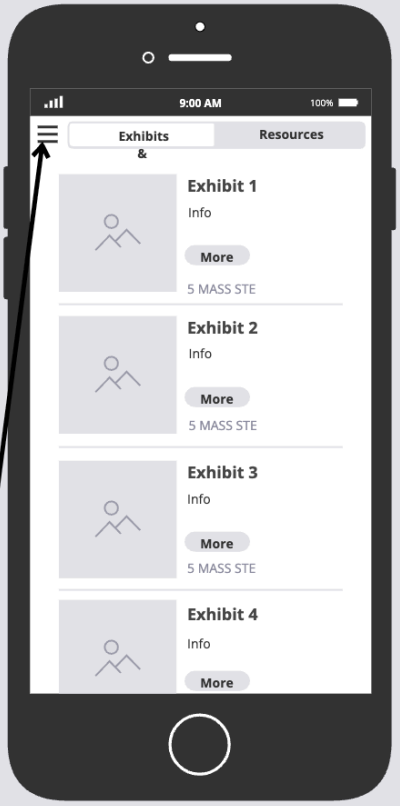
Map



Dashboard (Home Page)



Educator Search Result Page



Link to dashboard in menu with other top level links

8. Mood board: description of images and image of mood board

Fonts

Cascadia Code

Aa Bb Cc Dd Ee
Ff Gg Hh Ii Jj
Kk Ll Mm Nn Oo
Pp Qq Rr Ss Tt
Uu Vv Ww Xx Yy
Zz 0 1 2 3 4 5
6 7 8 9 ! ?

Verdana

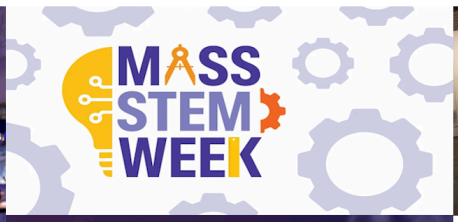
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Ff Gg Hh Ii Jj
Kk Ll Mm Nn Oo
Pp Qq Rr Ss Tt
Uu Vv Ww Xx Yy
Zz 0 1 2 3 4 5
6 7 8 9 ! ?

Colors

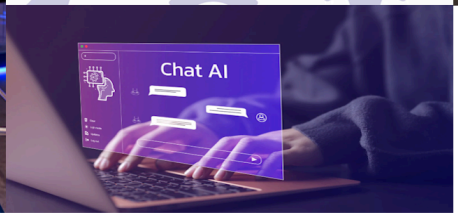
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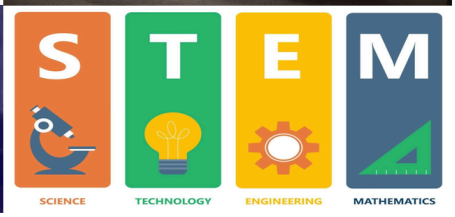
Example Images











[Leave the following section blank when you submit Assignments 1 & 2. You will need to fill this for Assignment 4.]

Assignment 4: Usability Evaluation Plan and Final Website

1. Usability Testing Plan

a. **Scope**

b. **Purpose/Questions**

c. **Schedule/Location**

d. **Sessions**

e. **Equipment**

f. **Participants**

g. **Task/Scenarios and Reasons (3-6)**

h. **Metrics/Questions**

i. **Roles**

2. URL/URI of your website (e.g. http://web.simmons.edu/~username)

3. URL/URI of AI-generated version (enter this only after the peer review process is completed)

4. Comments: 1-2 sentences explaining success with validation and any other comments pertaining to the website.